

AI Security Red Team Assessment

Adversarial Attacks, API Vulnerabilities, and Defenses

A technical evaluation of a locally deployed CIFAR-10 image classification model and FastAPI endpoint under simulated adversary conditions.

TARGET SYSTEM

PyTorch Simple CNN + FastAPI

TESTING SCOPE

Local Deployment (127.0.0.1:8000)

EVALUATION FOCUS

Recon → Evasion → Defense →
Re-Test

Assessment Scope & Methodology

01. OVERVIEW



🎯 Core Objectives

API Reconnaissance: Identify endpoint exposures, information disclosure, and input handling flaws.

Adversarial Evasion: Quantify robustness degradation under FGSM and PGD gradient attacks.

Hardening & Defense: Apply adversarial training and evaluate quantitative attack success reduction.

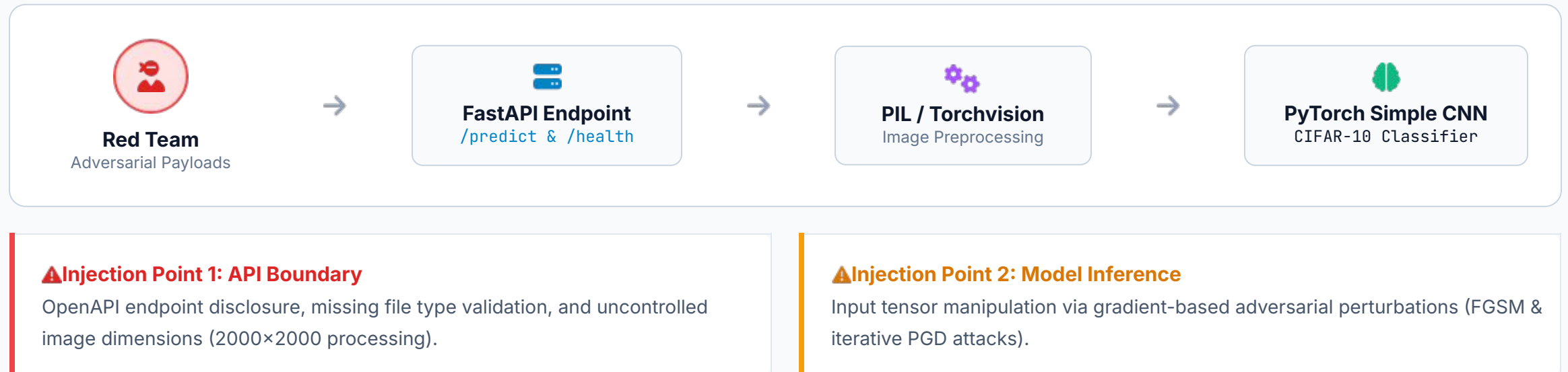
🔒 Strict Testing Boundaries

Target Scope: Restricted strictly to locally hosted CIFAR-10 model and FastAPI application.

Environment: `http://127.0.0.1:8000`

Ethical Isolation: Zero external network calls or third-party infrastructure targeted.

Target System Architecture & Attack Vectors



Target Model & Clean Baseline

03. BASELINE

Model Pipeline: `cifar10-simple-cnn-pytorch`

```
Input: CIFAR-10 Image (32×32×3)
→ Conv1: 3 → 32 channels
→ Conv2: 32 → 64 channels
→ Conv3: 64 → 128 channels
→ FC1: 2048 → 512 neurons
→ FC2: 512 → 10 classes
```

Classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck

62.38%

CLEAN TEST ACCURACY

Evaluated on uncorrupted CIFAR-10 test dataset.

SAMPLE BASELINE INFERENCE

True Class	→	Prediction	Confidence
cat		cat	68.01%

Reconnaissance & API Surface Findings

Finding	Evidence / Payload	Security Impact & Risk
OpenAPI Exposure	/results/openapi.json	Exposes endpoint schemas and payload definitions, enabling precise adversary reconnaissance.
Server Disclosure	Server: Uvicorn	Discloses underlying application stack, aiding targeted HTTP vulnerability discovery.
Unhandled Invalid File	test.txt → HTTP 500	Triggers PIL.UnidentifiedImageError. Unsanitized 500 error risks internal logic disclosure.
Missing File Validation	Empty body → HTTP 422	Unvalidated input boundary accepts malformed HTTP requests prior to image verification.
Large Image Acceptance	2000×2000 image accepted	No resource caps or dimension enforcement at API boundary; vulnerability to denial-of-resource.

Fast Gradient Sign Method (FGSM) Attack

ATTACK CONFIGURATION

$\epsilon = 0.03$

Single-step perturbation along the gradient sign vector to maximize loss.

SUCCESSFUL EVASION EXAMPLE

True Class
cat



FGSM Prediction
frog

FGSM Epsilon vs. Attack Success Rate (ASR)

Epsilon (ϵ)	Attack Success Rate (ASR)
0.00	0.00%
0.01	25.97%
0.03 (Test)	60.15%
0.05	77.11%
0.10	92.34%

Projected Gradient Descent (PGD) Attack

Iterative Attack Parameters

Perturbation Bound (ϵ)	0.03
Step Size (α)	0.005
Iterations (Steps)	10 steps

>_ End-to-End API Exploit Verification

PGD adversarial sample transmitted directly via REST payload to FastAPI endpoint (/predict):

```
{
  "filename": "pgd_adversarial_cat.png",
  "predicted_class": "frog",
  "confidence": 0.7708
}
```

✅ Standalone evasion confirmed live through API serving stack.

Adversarial Defense & Quantitative Results

Model Version	Attacked Samples	PGD Attack Success Rate	Accuracy Under PGD
Original Model	6,114	79.56%	20.44%
FGSM-Defended Model	6,953	53.49%	46.51%
PGD-Defended Model	7,149	43.07%	56.93%

KEY QUANTITATIVE FINDING

36.49 Percentage Point Reduction in Attack Success Rate

79.56% → 43.07%

Precision Note: PGD adversarial training significantly improved robustness, but did not eliminate vulnerability completely.

Mitigations & Industry Framework Mapping

Model Layer Mitigations

PGD adversarial training, continuous perturbation benchmarking, and safety evaluation loops.

API Layer Mitigations

Strict image dimension bounds, MIME type checking, generic error handling, and rate limits.

Model Integrity Controls

Enforce SHA-256 weight hash validation before loading model files into memory.

Framework Mapping

MITRE ATLAS

- Adversarial ML / Evasion
- Input Manipulation & Resource Abuse
- Information Disclosure

NIST AI RMF

- Security & Resilience Pillar
- MAP & MEASURE Functions

Assessment Conclusion & Key Takeaways

"The assessment demonstrated both ML-level and API-level weaknesses. Adversarial training improved robustness against the evaluated PGD attack (reducing attack success by 36.49 percentage points), while API validation and resource controls are required to strengthen the serving layer."

Thank You

Open for Questions & Live Defense Evaluation

Target: CIFAR-10 Simple CNN & FastAPI